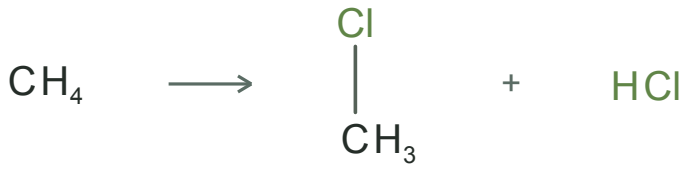
Alkanes and hydrogenation

Choose one complete row. + means select together. None is the dropdown default.

Methane substitution



methane chloromethane hydrogen chloride

Reagent option(s)	Catalyst	Heating	Extra option
Cl_2	None	None	UV light
Br_2	None	None	UV light

Example shows chlorine. Choose the matching halogen.

Cracking

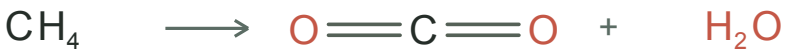


butane ethane ethene

Reagent option(s)	Catalyst	Heating	Extra option
—	Silica / alumina	Strong heat	—

Use the supplied coproduct to determine the other carbon chain.

Complete combustion



methane carbon dioxide water

Reagent option(s)	Catalyst	Heating	Extra option
Excess O_2	None	Ignition	—

Alkene → alkane



ethene ethane

Reagent option(s)	Catalyst	Heating	Extra option
H_2	Pt / Ni / Pd	None	—

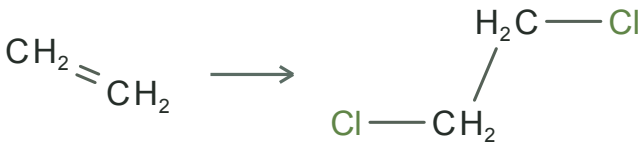
Product species shown; balancing coefficients and spectator salts omitted.

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Addition and substitution

Each row is one complete method. “or” means choose either heating option.

Halogen addition



ethene 1,2-dichloroethane

Reagent option(s)	Catalyst	Heating	Extra option
Cl_2 (organic solvent)	None	None	—
Br_2 (organic solvent)	None	None	—
I_2 (organic solvent)	None	None	—

Example shows chlorine. Br_2 gives the dibromo product; I_2 gives the diiodo product.

Hydrogen halide addition



prop-1-ene 2-chloropropane

Reagent option(s)	Catalyst	Heating	Extra option
HCl	None	None	—
HBr	None	None	—
HI	None	None	—

Example shows HCl. HBr / HI give the corresponding bromo / iodo product.

Haloalkane → alcohol

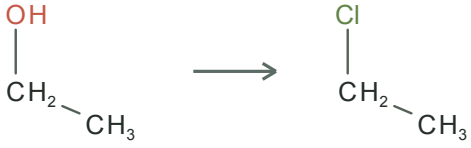


bromoethane ethanol

Reagent option(s)	Catalyst	Heating	Extra option
NaOH(aq)	None	Heat or Heat under reflux	—

Also applies to chloroalkanes and iodoalkanes.

Alcohol → haloalkane



ethanol chloroethane

Reagent option(s)	Catalyst	Heating	Extra option
HCl / PCl_3	None	None	—
HBr / PBr_3	None	None	—
HI	None	None	—

Example shows the chloro product. Choose the matching halogen option.

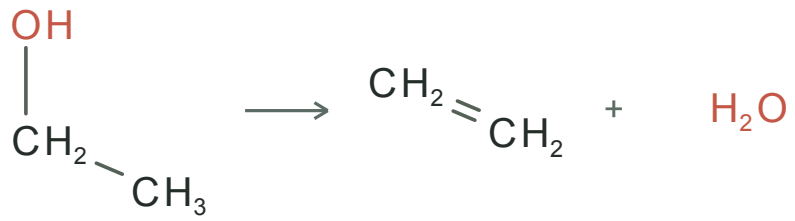
Product species shown; balancing coefficients and spectator salts omitted.

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Dehydration and oxidation

Each row is one complete method. “or” means choose either heating option.

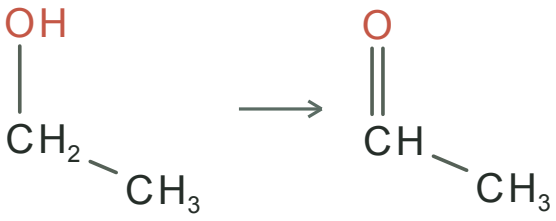
Alcohol → alkene



ethanol ethene water

Reagent option(s)	Catalyst	Heating	Extra option
Concentrated H_2SO_4	None	Heat	—
—	Al_2O_3	Heat	—

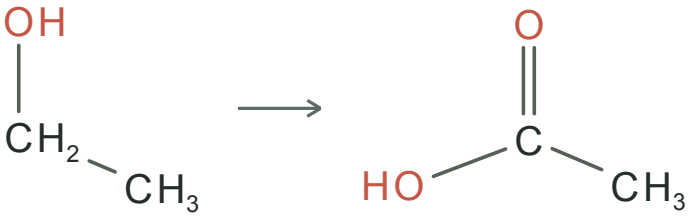
Primary alcohol → aldehyde



ethanol ethanal

Reagent option(s)	Catalyst	Heating	Extra option
$K_2Cr_2O_7(aq)$ / $H^+(aq)$	None	Heat and distil	—

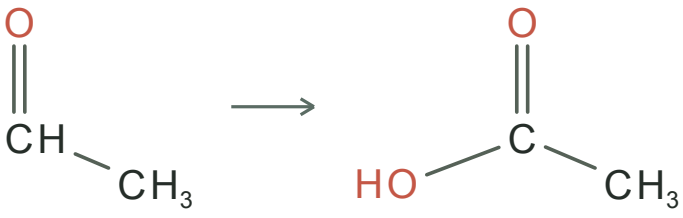
Primary alcohol → carboxylic acid



ethanol ethanoic acid

Reagent option(s)	Catalyst	Heating	Extra option
$K_2Cr_2O_7(aq)$ / $H^+(aq)$	None	Heat under reflux	—

Aldehyde → carboxylic acid



ethanal ethanoic acid

Reagent option(s)	Catalyst	Heating	Extra option
$K_2Cr_2O_7(aq)$ / $H^+(aq)$	None	Heat or Heat under reflux	—

Product species shown; balancing coefficients and spectator salts omitted.

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Ketones and reduction

Each row is one complete method. “or” means choose either heating option.

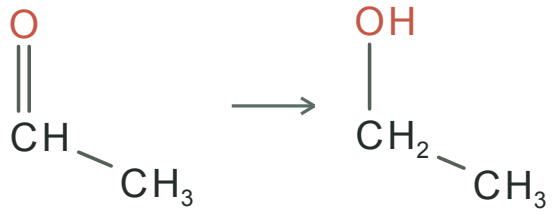
Secondary alcohol → ketone



propan-2-ol propan-2-one

Reagent option(s)	Catalyst	Heating	Extra option
$K_2Cr_2O_7$(aq) / H^+(aq)	None	Heat or Heat under reflux	—

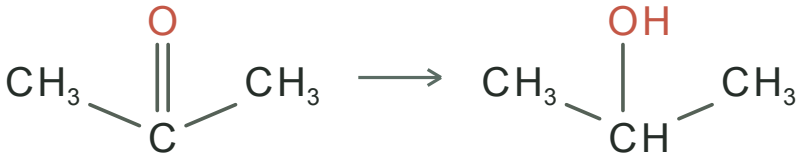
Aldehyde → primary alcohol



ethanal ethanol

Reagent option(s)	Catalyst	Heating	Extra option
$NaBH_4$(aq)	None	None	—
1. $LiAlH_4$, dry ether → 2. H^+(aq)	None	None	—

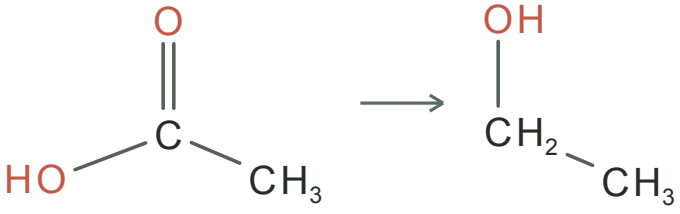
Ketone → secondary alcohol



propan-2-one propan-2-ol

Reagent option(s)	Catalyst	Heating	Extra option
$NaBH_4$(aq)	None	None	—
1. $LiAlH_4$, dry ether → 2. H^+(aq)	None	None	—

Carboxylic acid → primary alcohol



ethanoic acid ethanol

Reagent option(s)	Catalyst	Heating	Extra option
1. $LiAlH_4$, dry ether → 2. H^+(aq)	None	None	—

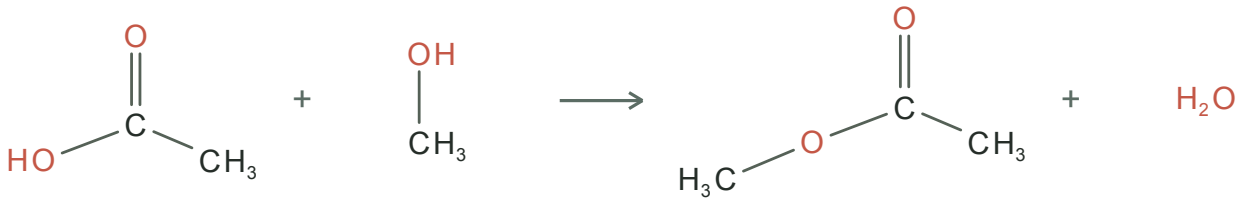
Product species shown; balancing coefficients and spectator salts omitted.

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Formation and ester hydrolysis

In synthesis, tick “Use in step” for the chosen alcohol. Single-reaction questions supply it.

Esterification

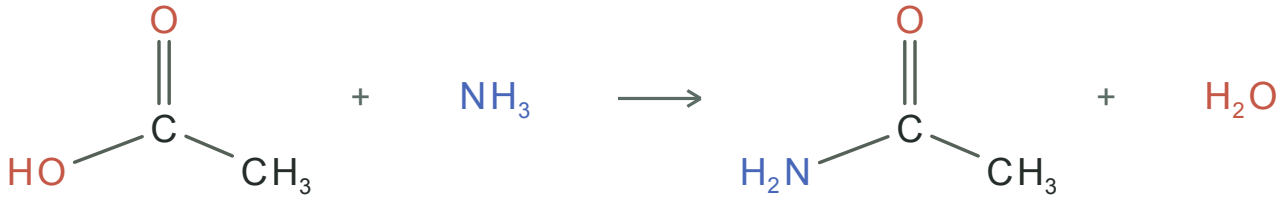


ethanoic acid — methanol — methyl ethanoate — water

Reagent option(s)	Catalyst	Heating	Extra option
methanol + Concentrated H_2SO_4	None	Heat or Heat under reflux	—
ethanol + Concentrated H_2SO_4	None	Heat or Heat under reflux	—
propan-1-ol + Concentrated H_2SO_4	None	Heat or Heat under reflux	—

Methanol → methyl ester; ethanol → ethyl ester; propan-1-ol → propyl ester.

Amide formation

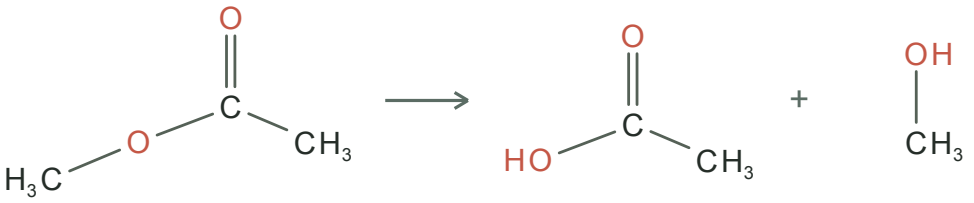


ethanoic acid — ammonia — ethanamide — water

Reagent option(s)	Catalyst	Heating	Extra option
NH_3	None	Heat	—

Select NH_3, then heat.

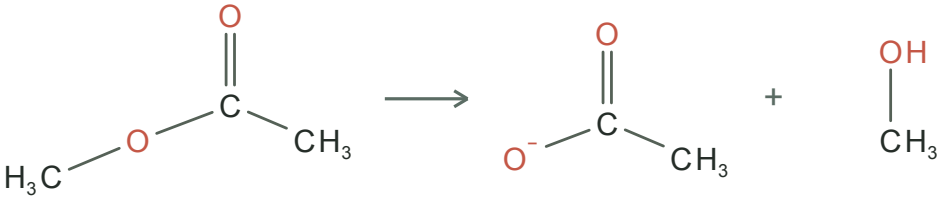
Ester hydrolysis · acidic



methyl ethanoate — ethanoic acid — methanol

Reagent option(s)	Catalyst	Heating	Extra option
Dilute H^+(aq)	None	Heat under reflux or Heat	—

Ester hydrolysis · alkaline



methyl ethanoate — ethanoate ion — methanol

Reagent option(s)	Catalyst	Heating	Extra option
NaOH(aq)	None	Heat under reflux or Heat	—

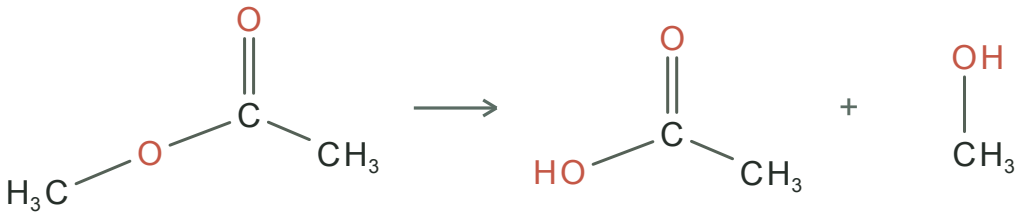
Product species shown; balancing coefficients and spectator salts omitted.

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Acid work-up and amide hydrolysis

Hydrolysis can assess either product. Draw the one that is not already supplied.

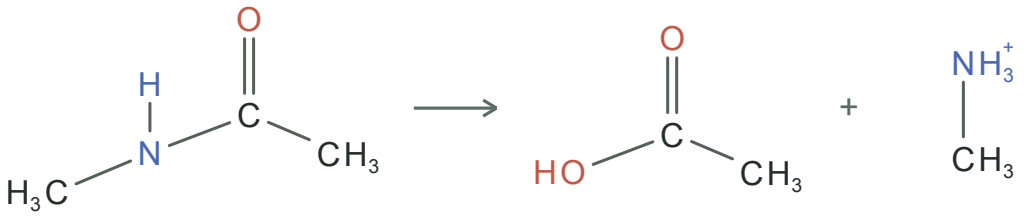
Ester hydrolysis · alkaline, then acid



methyl ethanoate | ethanoic acid | methanol

Reagent option(s)	Catalyst	Heating	Extra option
NaOH(aq)	None	Heat under reflux or Heat	Then H^+(aq)

Amide hydrolysis · acidic

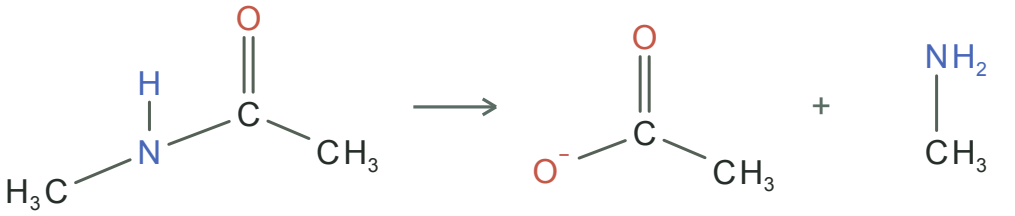


N-methylethanamide | ethanoic acid | methylammonium ion

Reagent option(s)	Catalyst	Heating	Extra option
Dilute H^+(aq)	None	Heat under reflux or Heat	—

Unsubstituted amide: $RCONH_2 \rightarrow RCOOH + NH_4^+$.

Amide hydrolysis · alkaline

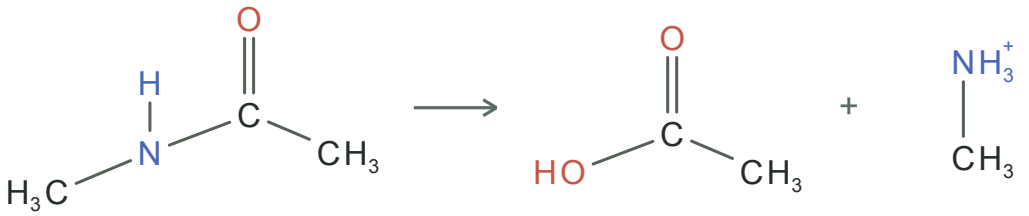


N-methylethanamide | ethanoate ion | methylamine

Reagent option(s)	Catalyst	Heating	Extra option
NaOH(aq)	None	Heat under reflux or Heat	—

Unsubstituted amide: $RCONH_2 \rightarrow RCOO^- + NH_3$.

Amide hydrolysis · alkaline, then acid



N-methylethanamide | ethanoic acid | methylammonium ion

Reagent option(s)	Catalyst	Heating	Extra option
NaOH(aq)	None	Heat under reflux or Heat	Then H^+(aq)

Unsubstituted amide: $RCONH_2 \rightarrow RCOOH + NH_4^+$ after excess acid work-up.

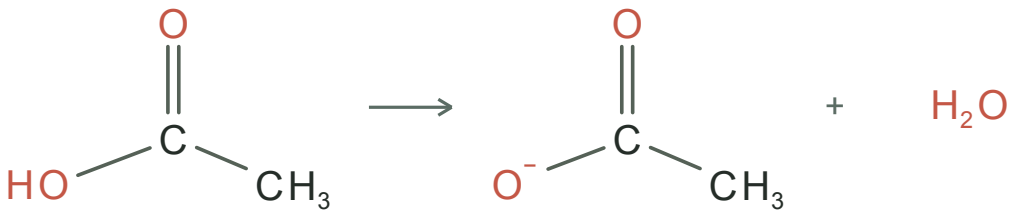
Product species shown; balancing coefficients and spectator salts omitted.

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Acid–base reactions

Each row is one complete method. “or” means choose either heating option.

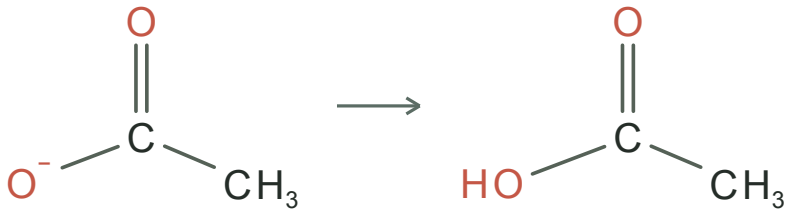
Carboxylic acid → carboxylate



ethanoic acid — ethanoate ion — water

Reagent option(s)	Catalyst	Heating	Extra option
NaOH(aq)	None	None	—

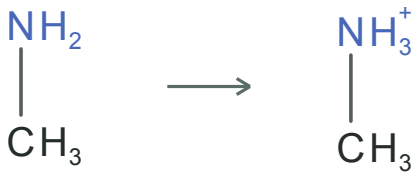
Carboxylate → carboxylic acid



ethanoate ion — ethanoic acid

Reagent option(s)	Catalyst	Heating	Extra option
Dilute H^+(aq)	None	None	—
HCl	None	None	—

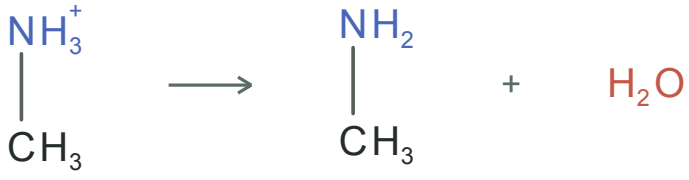
Amine → ammonium ion



methylamine — methylammonium ion

Reagent option(s)	Catalyst	Heating	Extra option
Dilute H^+(aq)	None	None	—
HCl	None	None	—

Ammonium ion → amine



methylammonium ion — methylamine — water

Reagent option(s)	Catalyst	Heating	Extra option
NaOH(aq)	None	None	—

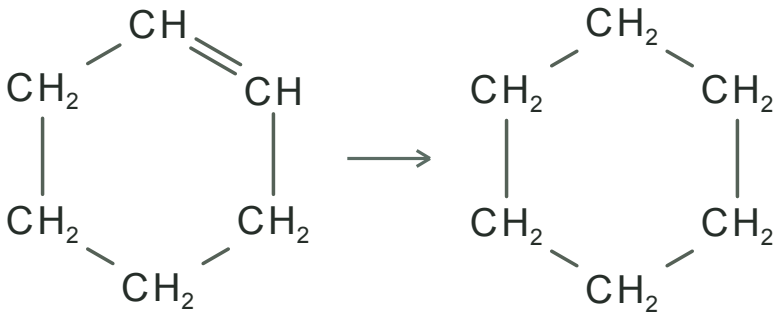
Product species shown; balancing coefficients and spectator salts omitted.

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Cyclic compounds · additions

The same options apply to the five- and six-membered rings generated by the game.

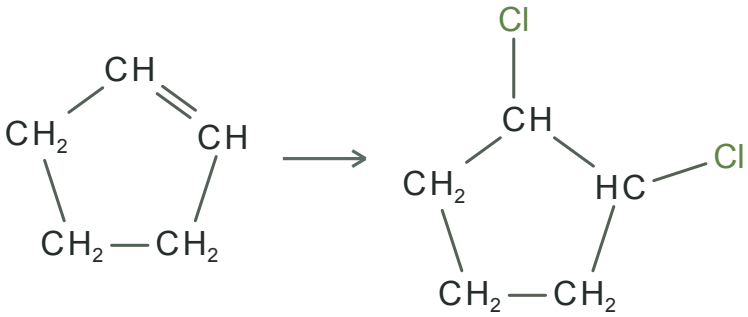
Cycloalkene → cycloalkane



cyclohexene — cyclohexane

Reagent option(s)	Catalyst	Heating	Extra option
H_2	Pt / Ni / Pd	None	—

Cycloalkene + halogen

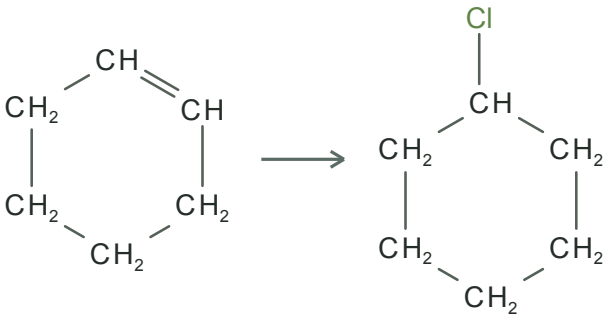


cyclopentene — 1,2-dichlorocyclopentane

Reagent option(s)	Catalyst	Heating	Extra option
Cl_2 (organic solvent)	None	None	—
Br_2 (organic solvent)	None	None	—
I_2 (organic solvent)	None	None	—

Example shows chlorine. Choose Br_2 / I_2 for the corresponding product.

Cycloalkene + hydrogen halide



cyclohexene — chlorocyclohexane

Reagent option(s)	Catalyst	Heating	Extra option
HCl	None	None	—
HBr	None	None	—
HI	None	None	—

Example shows HCl. Choose HBr / HI for the corresponding product.

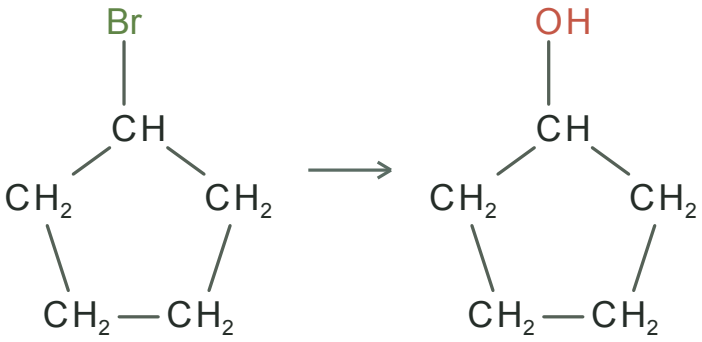
Product species shown; balancing coefficients and spectator salts omitted.

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Cyclic compounds · substitution and elimination

The same options apply to the five- and six-membered rings generated by the game.

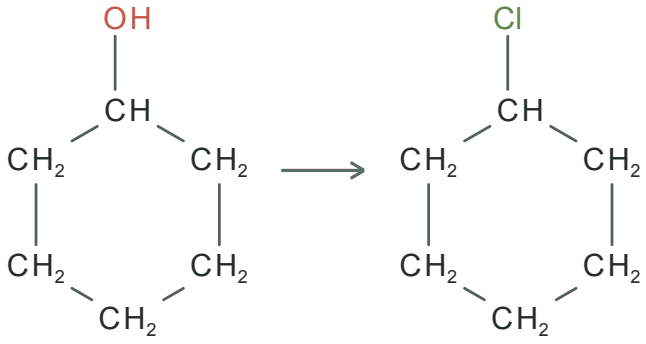
Halocycloalkane → cycloalkanol



bromocyclopentane — cyclopentanol

Reagent option(s)	Catalyst	Heating	Extra option
NaOH(aq)	None	Heat or Heat under reflux	—

Cycloalkanol → halocycloalkane

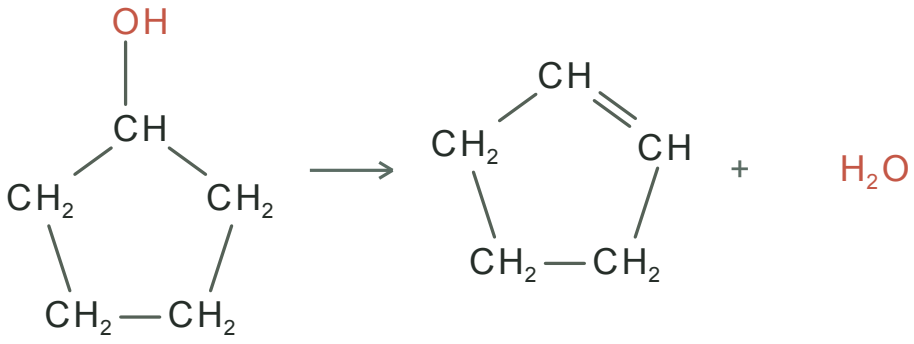


cyclohexanol — chlorocyclohexane

Reagent option(s)	Catalyst	Heating	Extra option
HCl / PCl_3	None	None	—
HBr / PBr_3	None	None	—
HI	None	None	—

Example shows chlorine. Choose the matching halogen option.

Cycloalkanol → cycloalkene



cyclopentanol — cyclopentene — water

Reagent option(s)	Catalyst	Heating	Extra option
Concentrated H_2SO_4	None	Heat	—
—	Al_2O_3	Heat	—

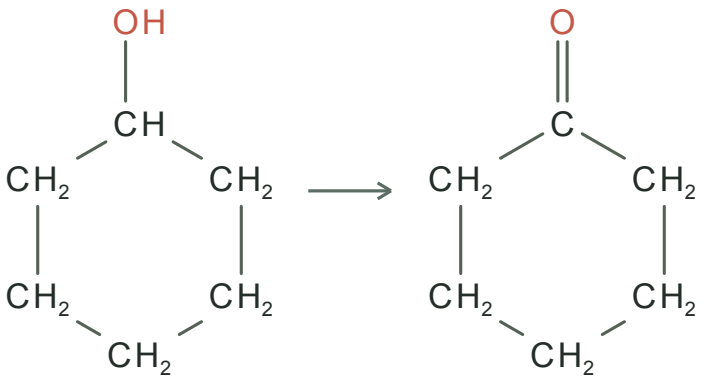
Product species shown; balancing coefficients and spectator salts omitted.

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Cyclic compounds · redox and combustion

The same options apply to the five- and six-membered rings generated by the game.

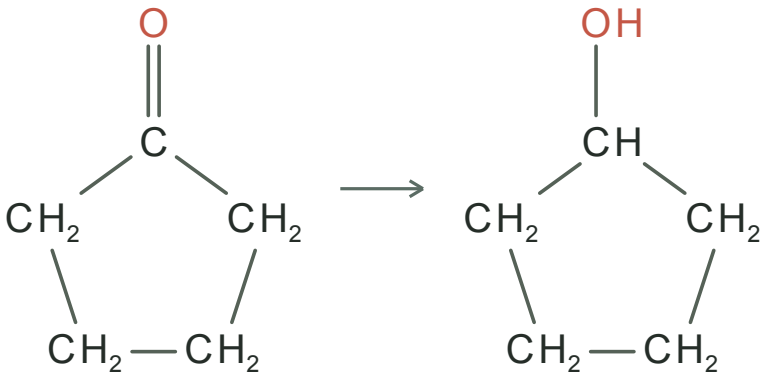
Cycloalkanol → cycloalkanone



cyclohexanol — cyclohexanone

Reagent option(s)	Catalyst	Heating	Extra option
$K_2Cr_2O_7$(aq) / H^+(aq)	None	Heat or Heat under reflux	—

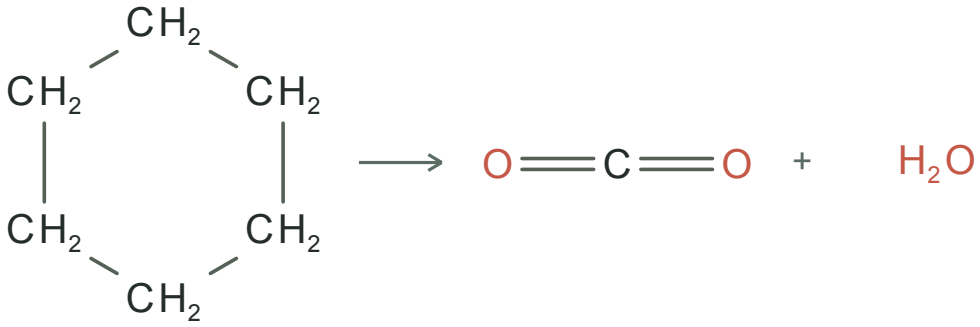
Cycloalkanone → cycloalkanol



cyclopentanone — cyclopentanol

Reagent option(s)	Catalyst	Heating	Extra option
$NaBH_4$(aq)	None	None	—
1. $LiAlH_4$, dry ether → 2. H^+(aq)	None	None	—

Cycloalkane combustion



cyclohexane — carbon dioxide — water

Reagent option(s)	Catalyst	Heating	Extra option
Excess O_2	None	Ignition	—

Product species shown; balancing coefficients and spectator salts omitted.

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